

Model Settings

t-SNE Perplexity



KNN Neighbors



Change the parameters above and the visualizations will update automatically.



Image Pattern Visualization using t-SNE

Apply t-SNE to visualize high-dimensional image datasets and identify hidden structures.

This project uses the **Digits dataset** to reduce high-dimensional image features into a 2-dimensional representation using PCA and t-SNE.



Dataset Information

Total Images

1797

Image Features

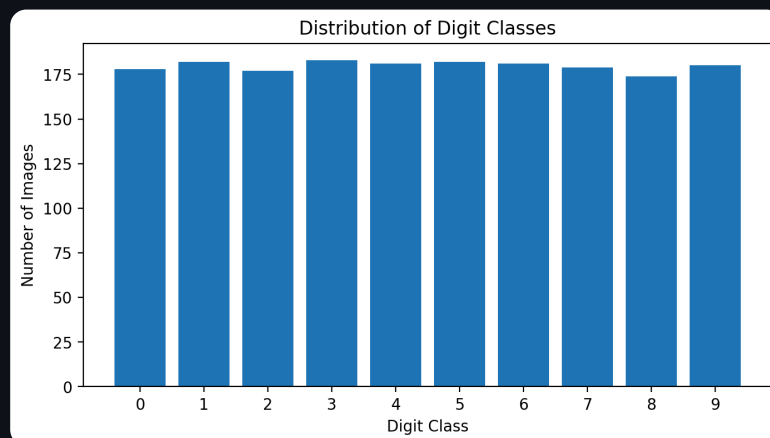
64

Number of Classes

10

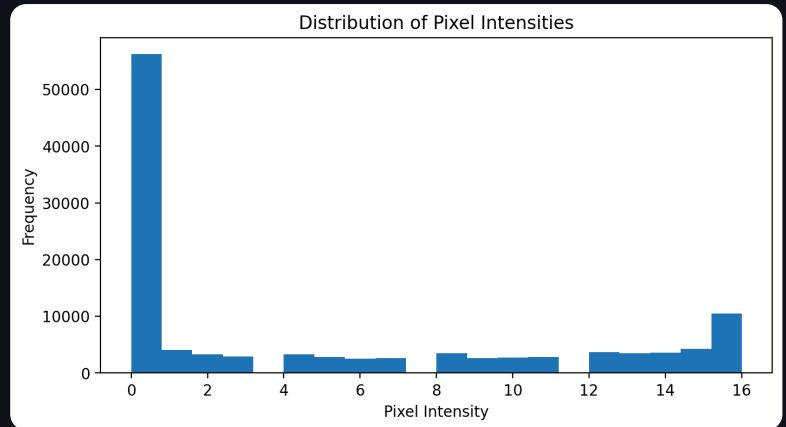


Class Distribution

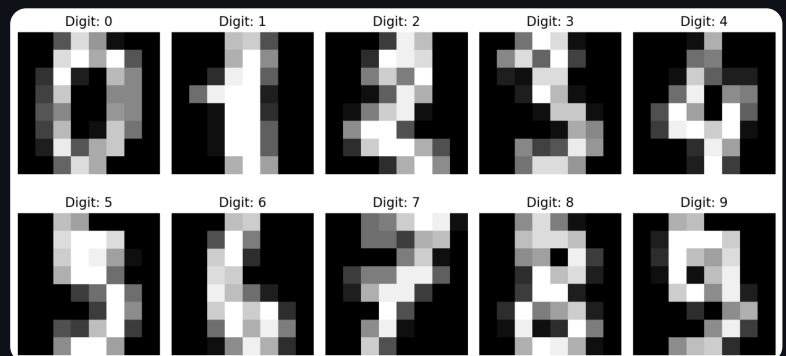


Pixel Intensity

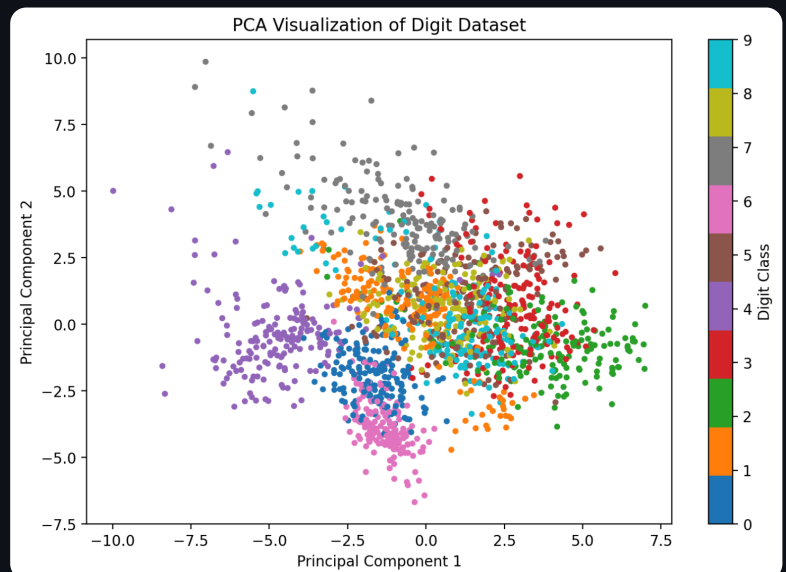
Distribution



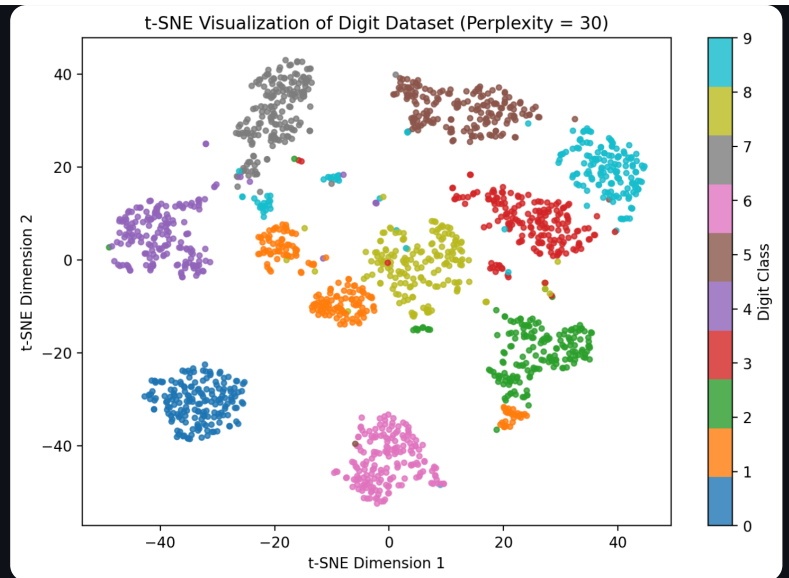
Sample Images



PCA Visualization



t-SNE Visualization



📊 KNN Classification

Accuracy	Precision	Recall	F1 Score
96....	96....	96....	96....

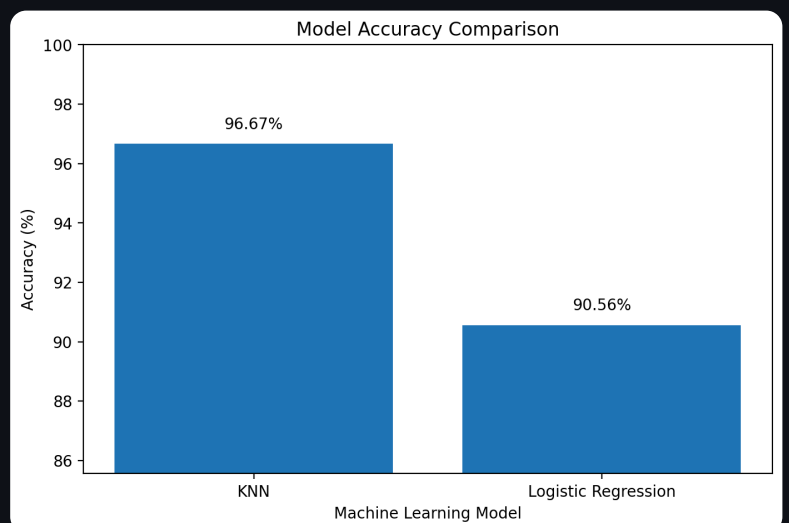


Logistic Regression

Accuracy	Precision	Recall	F1 Score
90....	90....	90....	90....

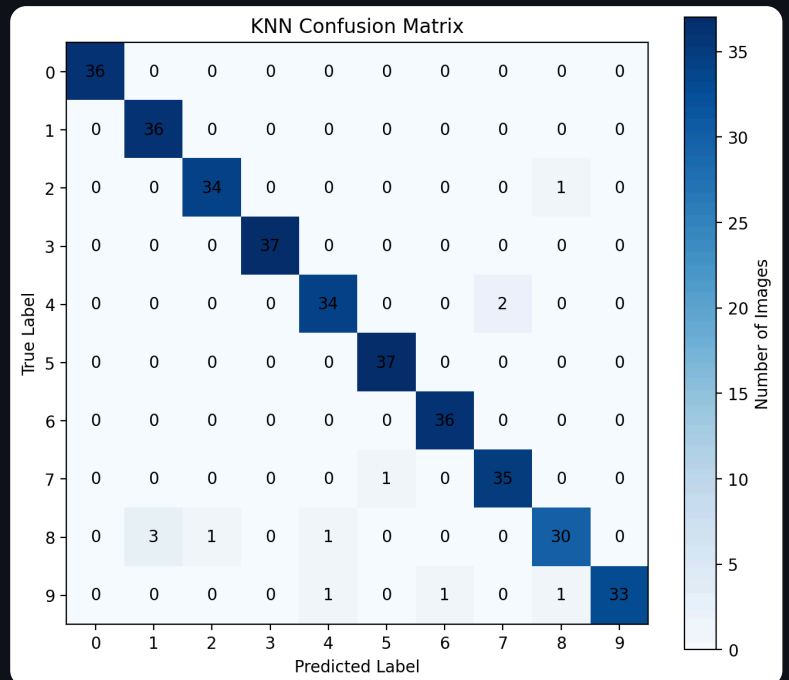


Model Comparison





KNN Confusion Matrix



Final Result

Best performing model: KNN with an accuracy of 96.67%.

Conclusion

t-SNE successfully reduces the high-dimensional image features into two dimensions, making hidden patterns, similarities, and class clusters easier to visualize.

The reduced representation can also be used for classification. In this project, KNN and Logistic Regression were evaluated and compared.

The model with the highest accuracy provides the best classification performance on the test data.

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